Advanced Mathematical Techniques Worksheet 5
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The purpose of this worksheet is to give you some practice working with the duplication identity of the Gamma function and using it to determine the value of various integrals involving logarithms and square roots. You may use Mathematica or another computer algebra system to check your work, but please do all of your work by hand.

1. This problem concerns the use of the duplication identity of the Gamma function,
$$\Gamma\left(1/2+\varepsilon\right)=2^{-2\varepsilon}\sqrt{\pi}\,\frac{\Gamma(1+2\varepsilon)}{\Gamma(1+\varepsilon)},$$
to investigate the behavior of the Gamma function in the vicinity of $1/2$.
   (a) Plug in $\varepsilon=4$ and work out the two sides of this identity separately. Do NOT actually multiply the numbers together!! Just write the necessary Gamma functions out, use the recurrence relation to bring the argument of the Gamma functions down to either 1 or $1/2$, and cancel everything you can. The purpose of this part is to give you a sense of *why* this identity is true.

   (b) Thinking of ε as a very small number, expand the right hand side of this identity. Keep up to ε^2. Note that this really isn't that bad if you combine all of the exponentials *before* expanding.

   (c) Your result from part (b) contains the first few terms in a Maclaurin expansion for $\Gamma\left(1/2+\varepsilon\right)$. Use this fact to determine the exact values of $\Gamma'\left(1/2\right)$ and $\Gamma''\left(1/2\right)$.

   (d) Use the recurrence relation in conjunction with your result from part (c) to determine the exact values of $\Gamma'\left(3/2\right)$ and $\Gamma''\left(3/2\right)$.

2. Use the duplication identity along with your expression from part (b) of problem 1 to determine the exact value of the following integrals. If an integral does not converge, indicate why it doesn't and move on. There is no need to determine the value of an integral that does not have a value!

   (a) $\int_0^\infty \sqrt{x}\,\ln x\, e^{-4x}dx$. (b) $\int_0^\infty \frac{\ln x}{x\sqrt{x}}e^{-2x}dx$. (c) $\int_0^\infty \frac{\ln^2 x}{\sqrt{x}}e^{-x}dx$.

3. Use the duplication identity along with your expression from part (b) of problem 1 and the Beta function integral representation of the first kind to determine the exact value of the following integrals.

(a) $\int_0^1 t \ln t \sqrt{1-t}\, dt$.

(b) $\int_0^1 \sqrt{\frac{t}{1-t}} \ln t\, dt$.

(c) $\int_0^1 \frac{\ln t}{\sqrt{1-t^2}} dt$.

(d) $\int_0^1 \frac{\ln^2 t}{\sqrt{1-t^2}} dt$.

4. Use the expressions derived in class for the Beta function integral representation of the second kind to determine the value of the following integrals. Note that these integrals do not require the introduction of the parameter ε or any expansion. If an integral does not converge, indicate why it doesn't and move on. There is no need to determine the exact value of an integral that does not have a value!

(a) $\int_0^\infty \frac{u \sqrt[3]{u}}{(u+1)^4} du$.

(b) $\int_0^\infty \frac{\sqrt{u} \ln u}{u^3+1} du$.

(c) $\int_0^\infty \frac{u^2 \sqrt{u} \ln^2 u}{u^3+1} du$.